

An Assessment of Faunal Diversity and its Conservation in Shipwrecks in Indian Seas

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INTRODUCTION

Coral reefs constitute a diverse and vulnerable ecosystem, characterized by a complex interdependence of plants and animals. These are massive limestone structures built up through the constructional cementing processes and depositional activities of animals of the class Anthozoa (Order: Scleractinia) as well as all other calcium carbonate secreting animals and calcifying algae (Venkataraman, 2000; Geetha and Kumar, 2012). Nowadays, worldwide coral reef ecosystems are declining continuously due to overfishing, pollution and other anthropogenic and natural disturbances (Venkataraman and Rajan, 1998; Venkataraman, 2000, 2006; Yogesh Kumar and Geetha, 2012 a,b,c; Kumar *et al.*, 2014). Artificial reefs and restoration are considered to be one of the major coral reef conservation techniques to mitigate the negative effects of anthropogenic activities on the reef ecosystem (Yogesh Kumar and Geetha, 2012c).

Sunken shipwrecks and man-made submerged structures may be called artificial reefs (Yan and Yan, 2003; Yogesh Kumar and Raghunathan, 2012). Shipwrecks offer substrata for settlement of fish and fouling organisms. This kind of artificial reef is common around the world, and artificial reefs have been used to increase fish production, to improve ecological and economic functions, to control beach erosion and trawl fishing and to conserve biodiversity (Seaman and Jensen, 2000; Baine, 2001; Perkol-Finkel and Benayahu, 2004, 2005). Although numerous studies have focused on artificial reefs, there are still many research gaps regarding their performance and potential applications (Svane and Peterson, 2001). The aim of the present study was to take a census of the faunal communities in five different shipwreck artificial reefs in Indian seas.

Specifically, the authors compared the benthic communities of recent and old shipwrecks and examined possible correlations between data from vertical and horizontal portions in shipwrecks.

METHODS

The studies were conducted on the Vembar wreck (Gulf of Mannar), Zuis wreck (Goa), North Bay wreck (Port Blair, South Andaman), Peel wreck (Havelock, South Andaman) and Japan wreck (Car Nicobar). The coordinates, dimensions and depth of all the wrecks are presented in [Table 25.1](#). The census of the benthic communities was made on the vertical and horizontal surfaces. The percentage of cover was recorded for hard corals, soft corals, sponges, tunicates, molluscs and algae in each shipwreck in 2 m long and 0.1 m wide flexible fibreglass measuring tape transects with the help of SCUBA, following line intercept transect and belt transect methods described in [Perkol-Finkel and Benayahu \(2004\)](#). To assess the fish diversity by visual census in each shipwreck, diversity and abundance of reef fish within 2.5 m on either side of the transect line were noted ([English *et al.*, 1997](#)). An underwater camera was used for video documentation for further studies.

Biodiversity indices such as species richness, evenness and Shannon-Wiener diversity index were compared among the shipwrecks. Margalef's species richness method was used for calculating species richness.

RESULTS

Transect surveys were conducted periodically from 2007 to 2014. Fouling invertebrates, including live corals and soft corals, sponges, ascidians, bryozoans, polychaetes, nudibranchia, barnacles, bivalves, molluscs, sea urchins, algae and other faunal communities ([Figures 25.1 and 25.2](#)) that intercepted the transects were recorded and the projected length of each on the transects was measured. Quantitative data on each benthic life form were recorded from each shipwreck and are presented in [Table 25.2](#). Live coral cover on the vertical surfaces was calculated to be 6.5, 5.6, 3.7 and 3.7 percent at the North Bay, Zuis, Vembar and Japan wrecks, respectively. Live coral cover on the horizontal surfaces was 26.1, 24.9, 24.7, 21.5 and 9.5 percent at the Japan, Vembar, Zuis, North Bay and Peel wrecks, respectively. The Japan wreck exhibited the greatest live coral cover, while the Peel wreck exhibited the least live coral cover and soft coral cover. Generally, massive coral dominated compared with the *Acropora* branching corals.

Soft coral and barnacle were predominantly noted on the vertical surfaces compared with the horizontal. Maximum cover of sponges was reported for both orientations; other faunal communities produced a far smaller contribution to the fouling organisms ([Table 25.2](#)). The faunal community was identified underwater—mostly to species level, and when this was not possible they were assigned to genera and family.

The percentage of fish abundance ([Figure 25.3](#)) was assessed by family among the five shipwrecks ([Table 25.3](#)). The fish of the Lutjanidae, Pempheriidae,

TABLE 25.1 Features of the Studied Shipwrecks

	Vembar Wreck	Zusis Wreck	North Bay Wreck	Peel Wreck	Japan Wreck
Coordinates (latitude; longitude)	09°02'08.20"N; 78°24'03.02"E	15°21'00.43"N; 73°46'46.68"E	11°43'00.56"N; 92°45'60.60"E	12°03'84.20"N; 92°57'81.10"E	09°10'88.30"N; 92°50'12.30"E
Approx. age (year)	25–30	40	30–40	8–10	40–50
Place	Gulf of Mannar	Goa	Port Blair	Havelock	Car Nicobar
Dimensions (m): length × width × height	51 × 11 × 9	72 × 14 × 9	68 × 14 × 9	21 × 8 × 6	70 × 13 × 9
Depth (m)	9	9–11	10	9–12	28
No. of horizontal transects	25	20	20	10	15
No. of vertical transects	10	18	15	8	10
Present status	Fishing ground	Recreational diving	Fishing and recreational diving	Recreational diving	Fishing ground

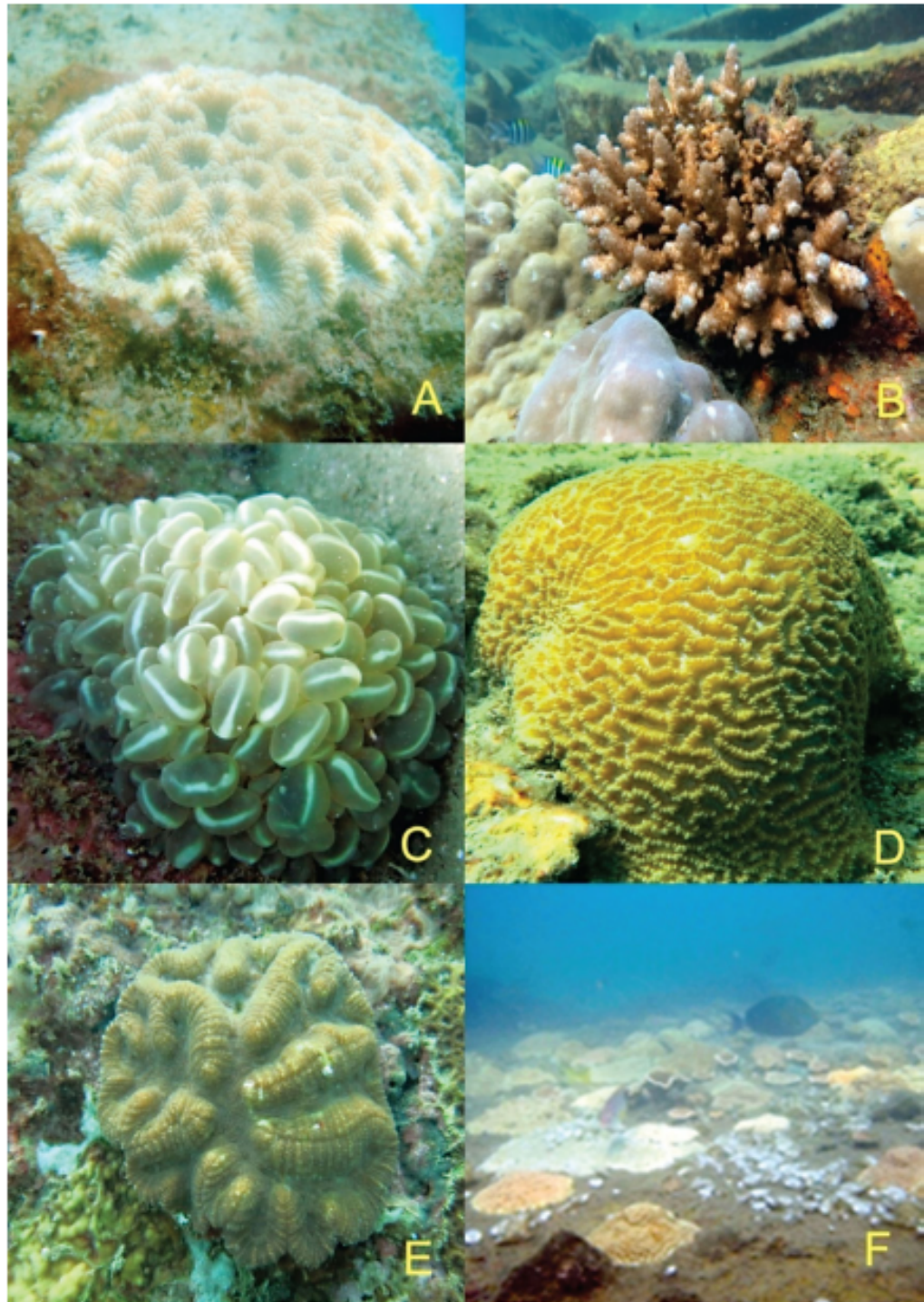


FIGURE 25.1 Live corals on shipwreck. (A) *Favia fava*; (B) *Acropora hemprichii*; (C) *Pleurogyra sinulosa*; (D) *Platygyra lamellina*; (E) *Symphyllia recta*; (F) Different types of coral on horizontal surface. (Please see color plate at the back of the book.)

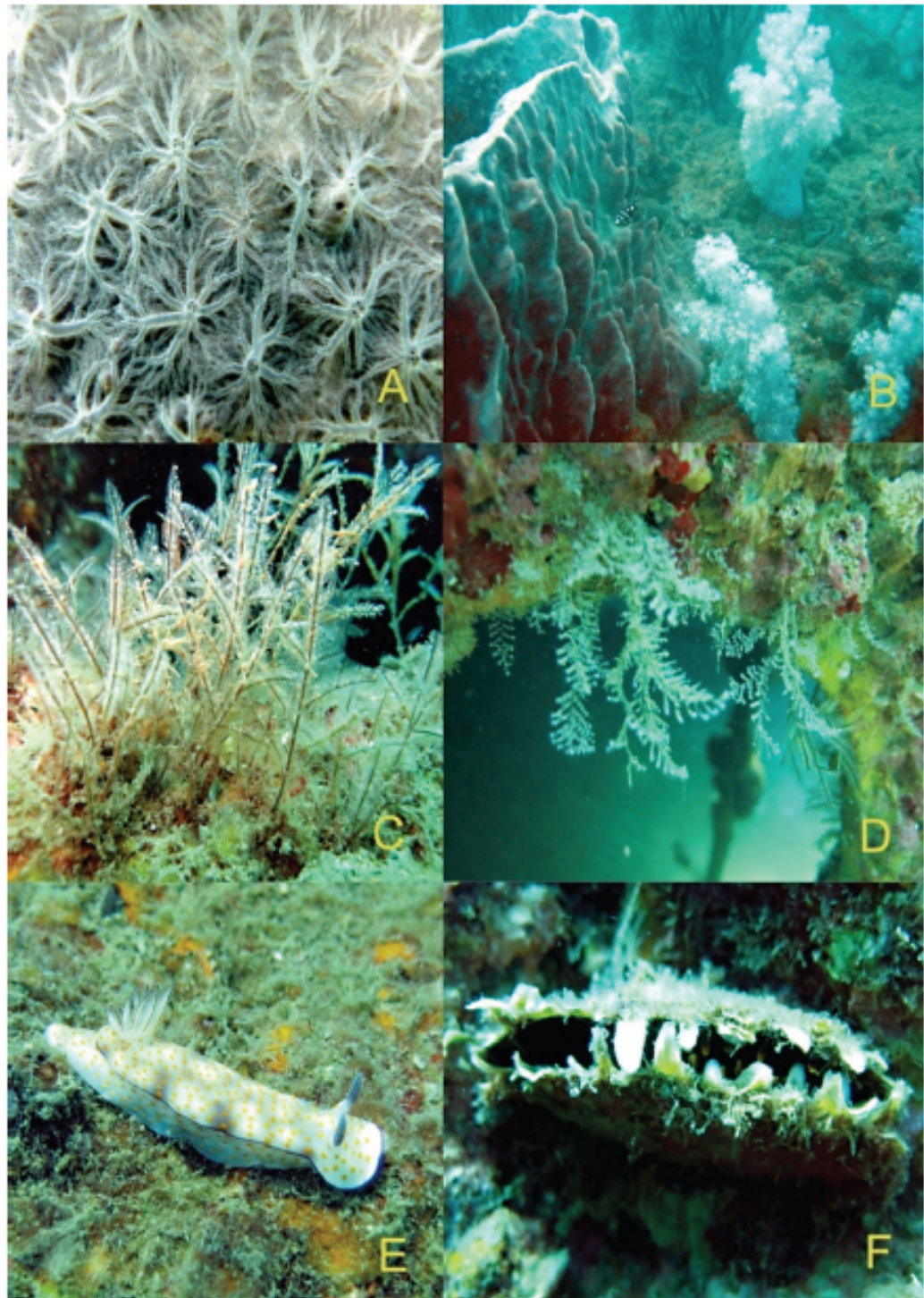


FIGURE 25.2 Other reef-associated fauna on shipwreck. (A) Sponges; (B) Nephtheidae (soft corals); (C) *Macrobynchia philippina* (hydrozoans); (D) *Dynamena*; (E) *Risbecia ghardaqana* (nudibranchs); (F) *Pinctada margarifera* (bivalve molluscs). (Please see color plate at the back of the book.)

TABLE 25.2 Percentage Benthic invertebrate diversity for Each Shipwreck

	Vembar wreck		Zuis wreck		North Bay wreck		Peel wreck		Japan wreck	
	V	H	V	H	V	H	V	H	V	H
Live coral	3.7	24.9	5.6	24.7	6.5	21.5	0.0	9.5	3.7	26.1
Soft coral	18.7	1.1	20.9	6.5	23.4	14.2	10.9	8.8	23.8	4.5
Sponges	14.9	20.3	14.4	10.9	12.0	9.5	5.4	12.4	15.8	10.5
Ascidians	2.4	3.8	5.3	3.7	2.5	3.1	1.9	2.6	9.5	4.5
Bryozoans	13.5	8.7	4.3	10.4	4.5	8.9	12.3	17.5	3.6	9.7
Polychaetes	1.2	2.6	2.0	4.4	0.0	4.0	1.7	0.0	2.7	4.1
Nudibranchia	1.3	1.1	2.4	1.8	2.5	1.6	0.0	2.6	1.9	1.7
Barnacles	24.5	11.7	24.5	8.6	27.0	9.9	40.7	11.6	14.9	12.3
Bivalves and molluscs	4.9	14.7	8.7	10.7	9.1	9.6	10.9	13.2	10.2	11.6
Sea urchins	1.3	0.0	2.4	2.7	2.5	2.4	0.0	2.9	0.8	2.6
Algae	1.2	1.3	2.0	5.9	2.0	4.9	3.6	4.7	3.7	3.9
Others	12.5	9.6	7.7	9.8	8.0	10.3	12.7	14.3	9.3	8.3

H, horizontal; V, vertical.

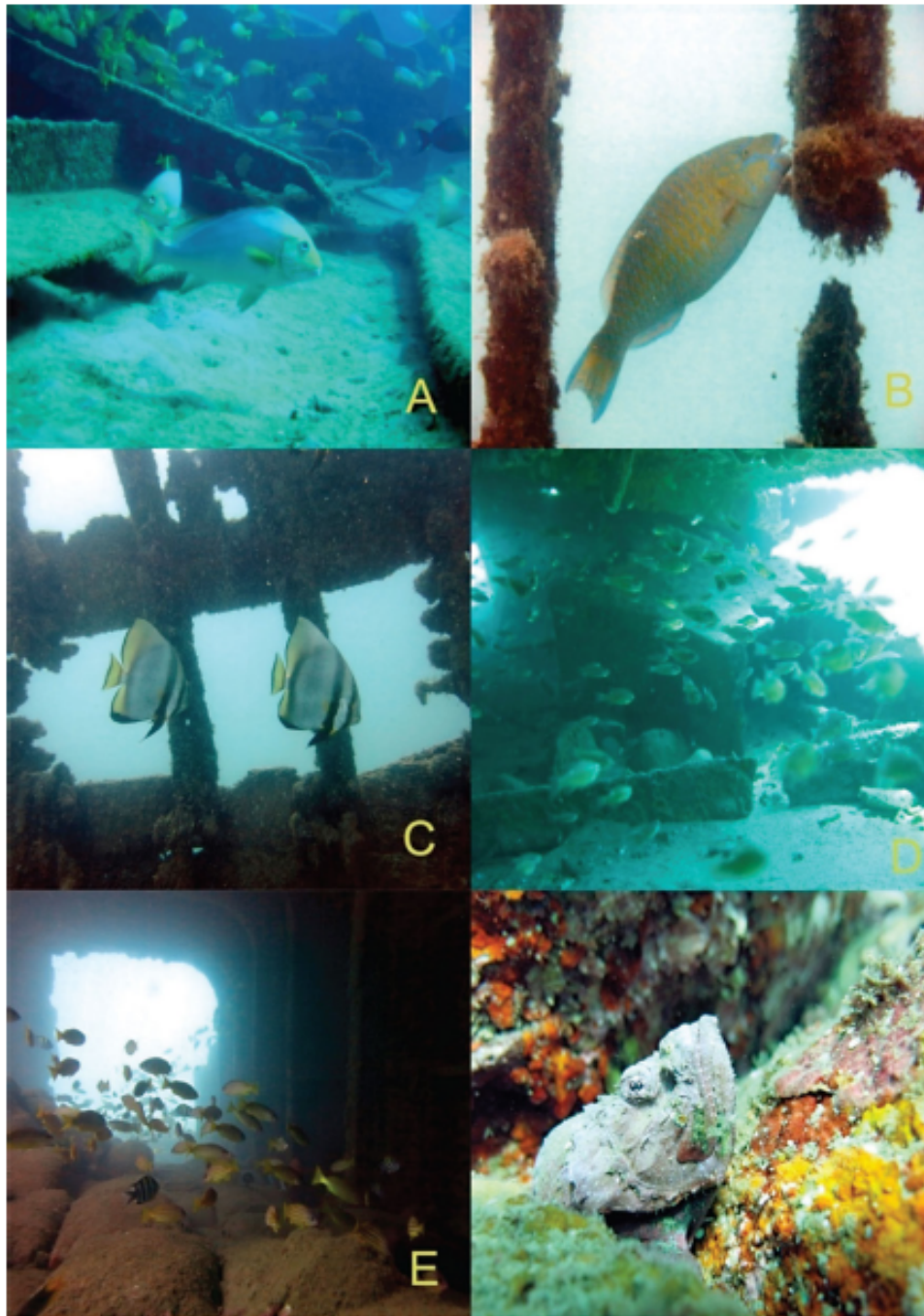


FIGURE 25.3 Reef-associated fish in shipwrecks. (A) *Plectorhinchus multivittatum*; (B) *Scarus psittacus*; (C) *Platax pinnatus*; (D) *Apogon* sp.; (E) Lethrinidae; (F) Scorpaenidae.

TABLE 25.3 Percentage Fish Diversity for Each Shipwreck

Sl. No.	Family	Common name	Vembar wreck	Zuisi wreck	North Bay wreck	Peel wreck	Japan wreck
1	Apogoniidae	Cadinalfish	11.6	8.5	7.8	16.3	14.6
2	Acanthuridae	Surgeonfish	1.2	0.8	1.9	3.3	1.8
3	Carangidae	Jackfish	7.0	4.6	4.7	0.0	7.3
4	Chaetodontidae	Butterflyfish	7.0	5.8	5.6	6.5	3.5
5	Cirrhitidae	Hawkfish	1.2	2.3	2.8	1.1	1.2
6	Ephippidiidae	Batfish	1.2	2.7	2.5	2.2	2.3
7	Balistidae	Triggerfish	2.9	5.4	3.8	8.7	0.9
8	Blenniidae	Blennies	0.0	0.8	1.6	1.1	2.0
9	Diodontidae	Porcupinefish	0.6	0.4	0.3	0.0	0.0
10	Gobiidae	Gobies	0.0	0.8	1.6	0.0	1.8
11	Labridae	Wrasses	12.2	5.8	5.6	5.4	4.4
12	Lutjanidae	Snappers	23.3	19.3	25.1	21.7	35.1
13	Muraenidae	Moray eel	1.2	1.5	0.3	0.0	0.3
14	Mullidae	Goatfish	2.3	3.1	1.9	5.4	2.9
15	Ostraciidae	Boxfish	1.2	1.9	1.3	1.1	1.2
16	Pempheridae	Sweepers	12.8	21.6	18.8	13.0	12.3
17	Pomacanthidae	Angelfish	1.2	1.5	2.5	2.2	2.3
18	Scaridae	Parrotfish	4.7	3.1	3.8	6.5	2.3
19	Scorpaenidae	Scorpionfish	4.1	4.6	3.1	2.2	0.9
20	Serranidae	Groupers	4.1	3.9	2.5	1.1	1.8
21	Siganidae	Rabbitfish	0.6	1.5	2.5	2.2	1.2

TABLE 25.4 Diversity Indices for Fish at Each Shipwreck

Parameter	Vembar wreck	Zusis wreck	North Bay wreck	Peel wreck	Japan wreck
No. of families	19	21	21	17	20
Total no of individuals	172	259	319	92	342
Diversity index, H'	2.45	2.56	2.53	2.4	2.27
Species richness, d	3.49	3.59	3.46	3.54	3.26
Species evenness, J	0.8	0.84	0.83	0.79	0.75

Apogoniidae and Labridae families dominated in all of the shipwrecks, followed by Carangidae, Chaetodontidae, Scaridae, Scorpaenidae, Serranidae, Balistidae, Mullidae, Pomacanthidae, Ostraciidae, Muraenidae, Ephippidae and Cirrhitidae.

The maximum number of individuals visually recorded was from the Japan wreck (342), followed by the North Bay wreck (319), Zusis wreck (259), Vembar wreck (172) and Peel wreck (92). The diversity indices Shannon-Wiener diversity index (H'), Pielou's evenness (J) and Margalef's species richness (d) were determined. H' was maximum for the Zusis wreck (2.56) and minimum for the Peel wreck (2.4); J ranged from 0.75 for the Japan wreck to 0.84 for the Zusis wreck; d was maximum (3.59) for the Zusis wreck and minimum (3.26) for the Japan wreck (Table 25.4).

DISCUSSION

Shipwrecks and other unplanned artificial reef structures offer a substratum for settlement of benthic invertebrates and fish. These structures mimic natural reef and attract a variety of fish, all kinds of marine life and recreational SCUBA divers (Black, 2001; Black and Mead, 2001; Sutton and Bushnell, 2007; Perkol-Finkel and Benayahu, 2004, 2009). The present surveys carried out at shipwrecks in Indian waters have revealed diversity and abundance of fauna including fish, live corals, soft corals, sponges, ascidians, bryozoans, polychaetes, nudibranchia, barnacles, bivalves, molluscs, sea urchins and algae. The live coral, soft coral and fish were more dominant in old shipwrecks (Japan, North Bay, Zusis and Vembar) than in the more recent shipwreck (Peel). More live corals were observed on horizontal surfaces than on vertical. But the percentage of soft coral was high for the vertical and low for the horizontal. This may be due to the environment, size, structure, geographical location, depth, surrounding substratum, type of materials and age (Baine, 2001; Rilov and Benayahu, 2000;

Sheng, 2000; Perkol-Finkel and Benayahu, 2005). Wrecks more than a hundred years old have been reported to have a hard coral cover similar to their adjacent natural reef (Riegl, 2001).

Artificial reefs attract reef fish. Therefore, consideration may be given to deploying such artificial reefs to increase fishery productivity (Einbinder *et al.*, 2006). Maximum abundance and distribution of reef fish at artificial reefs have been reported to be higher than in the surrounding coral reef environment (Wilhelmsson *et al.*, 1998). In the present study, we reported 21 families from the shipwrecks during the study period. Snapper (Lutjanidae) and grouper (Serranidae) were dominant among economically important and edible reef fish species at all the study station. This may be due to the availability of macro algae cover, fish young, nutrient richness and low disturbance (Williams and Polunin, 2001).

The present study locations are being actively used for recreational SCUBA diving and fishing. Therefore, fresh deployment of man-made artificial structures in many places on the ocean bottom could create new fertile habitats with higher biodiversity, and overexploitation caused by continuous fishing activities and trawler fishing could thus be reduced. Construction of artificial reefs in substantial numbers in near-shore areas, natural coral reefs would reduce the impact of anthropogenic activities on natural reefs. In this way, we can also protect and conserve these fragile ecosystems for future generations.

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